

Product Design Specification (PDS)

1. Project Title

- Product Name: **Water Rocket**
- Date: 23/05/25
- Prepared by: Group 3

2. Purpose / Background

This project involves designing and developing a water-powered rocket equipped with a Micro: bit controller. The goal is to measure key performance parameters such as maximum height and speed using onboard sensors. This educational product serves as a hands-on tool for learning principles of physics, programming, and engineering.

3. Scope

The rocket will be powered by pressurized water and constructed using a standard 2-liter plastic bottle. It will feature a 3D-printed PLA nose cone and fins for aerodynamics. A Micro: bit microcontroller will be integrated with sensors to measure and log height and speed during flight.

4. Target Market / Users

- Primary users: A-level student/ First year university students
- Secondary users: Hobbyists, science clubs
- Market segment: Educational kits
- User needs/expectations: Easy assembly, safe launch, measurable performance and reusability

5. Key Requirements

Requirement	Specification / Description
Functionality	Rocket must launch using the chosen () water pressure and record and transfer flight data remotely.
Performance	Achieve measurable flight with reliable height and speed readings.
Aesthetics	Compact, clean design with distinguishable components.
Ergonomics	Easy to assemble, operate and retrieve.
Dimensions	Standard 2-liter size bottle, fins and nose cone designed to fit appropriately

	(appendix 1)
Weight	352.64 g without adding any water
Materials	2-liter PET bottle, PLA for 3D printed parts, standard Micro: bit Board.
Compliance/Standards	Compliant with University safety standards for model rockets and launching procedure.
Cost	£10.74 cost of 3D printing (cost of R/D excluded)
Lifespan/Durability	Reusable for at least 10 launches with replaceable components.
Environmental Impact	Reuses/ Recycle plastic bottles; 3D printed PLA Recyclable.
Maintenance	Minimal – recharging battery and cleaning.
Safety	Pressure launches within chosen PSI limit, Rounded nose cone.
Power / Energy Source	Battery-powered Micro:Bit, water for propulsion.
Interfaces	USB for Micro:Bit programming. Data displayed on device/ computer

6. Constraints

Use of a 2-liter bottle, limited payload weight, budget constraints, maximum launch pressure of 10 PSI and time limits.

7. Assumptions

State any assumptions made about the user, environment, or usage.

8. Appendices

- Sketches



